ANSWER KEY

English Test

1. C	21. B	41. A	61. C
2. F	22. F	42. J	62. J
3. D	23. C	43. C	63. B
4. F	24. H	44. F	64. H
5. D	25. C	45. D	65. D
6. H	26. F	46. G	66. G
7. A	27. D	47. B	67. D
8. J	28. F	48. F	68. F
9. B	29. D	49. C	69. D
10. F	30. G	50. F	70. F
11. C	31. C	51. D	71. A
12. J	32. G	52. G	72. G
13. B	33. D	53. C	73. C
14. G	34. J	54. G	74. G
15. A	35. B	55. A	75. B
16. J	36. J	56. F	
17. B	37. B	57. A	
18. F	38. J	58. H	
19. D	39. D	59. C	
20. G	40. F	60. H	

Mathematics Test

1. D	21. A	41. B
2. H	22. K	42. H
3. A	23. C	43. A
4. H	24. K	44. J
5. D	25. D	45. B
6. K	26. G	46. K
7. C	27. A	47. D
8. J	28. F	48. J
9. C	29. E	49. D
10. J	30. G	50. G
11. D	31. D	51. B
12. J	32. F	52. J
13. E	33. B	53. E
14. H	34. H	54. J
15. D	35. E	55. E
16. J	36. G	56. J
17. C	37. C	57. A
18. J	38. J	58. K
19. B	39. C	59. D
20. G	40. G	60. G

Reading Test

1. B	21. C
2. F	22. J
3. D	23. A
4. G	24. F
5. B	25. D
6. H	26. G
7. A	27. D
8. J	28. J
9. C	29. A
10. F	30. H
11. A	31. D
12. H	32. F
13. D	33. C
14. F	34. J
15. B	35. C
16. J	36. H
17. A	37. B
18. F	38. J
19. D	39. B
20. H	40. H

Science Reasoning Test

1. A	21. A
2. H	22. J
3. D	23. B
4. G	24. G
5. A	25. B
6. H	26. C
7. B	27. A
8. H	28. H
9. C	29. C
10. F	30. J
11. C	31. B
12. F	32. A
13. B	33. A
14. J	34. F
15. A	35. A
16. H	36. H
17. B	37. C
18. G	38. F
19. C	39. C
20. J	40. H

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SCORING GUIDE

Your final reported score is your COMPOSITE SCORE. Your COMPOSITE SCORE is the average of all of your SCALE SCORES.

Your SCALE SCORES for the four multiple-choice sections are derived from the Scoring Table on the next page. Use your RAW SCORE, or the number of questions that you answered correctly for each section, to determine your SCALE SCORE. If you got a RAW SCORE of 60 on the English test, for example, you correctly answered 60 out of 75 questions.

Step 1 Determine your RAW SCORE for each of the four multiple-choice sections:

English _____

Mathematics _____

Reading _____

Science Reasoning _____

The following Raw Score Table shows the total possible points for each section.

RAW SCORE TABLE	
KNOWLEDGE AND SKILL AREAS	**RAW SCORES**
ENGLISH	75
MATHEMATICS	60
READING	40
SCIENCE REASONING	40
WRITING	12

Multiple-Choice Scoring Worksheet

Step 2 Determine your SCALE SCORE for each of the four multiple-choice sections using the following Scoring Worksheet. Each SCALE SCORE should be rounded to the nearest number according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32. If you answered 61 questions correctly on the English section, for example, your SCALE SCORE would be 29.

English	_____ RAW SCORE	× **36** =	_____	÷ **75** =	_____	**− 2**	(*correction factor)
					_____ SCALE SCORE		
Mathematics	_____ RAW SCORE	× **36** =	_____	÷ **60** =	_____	**+ 1**	(*correction factor)
					_____ SCALE SCORE		
Reading	_____ RAW SCORE	× **36** =	_____	÷ **40** =	_____	**+ 2**	(*correction factor)
					_____ SCALE SCORE		
Science Reasoning	_____ RAW SCORE	× **36** =	_____	÷ **40** =	_____	**+ 1.5**	(*correction factor)
					_____ SCALE SCORE		

*The correction factor is an approximation based on the average from several recent ACT tests. It is most valid for scores in the middle 50% (approximately 16–24 scale composite score) of the scoring range.

The scores are all approximate. Actual ACT scoring scales vary from one administration to the next based upon several factors.

If you take the optional Writing Test, you will need to combine your English and Writing scores to obtain your final COMPOSITE SCORE. Once you have determined a score for your essay out of 12 possible points, you will need to determine your ENGLISH/WRITING SCALE SCORE, using both your ENGLISH SCALE SCORE and your WRITING TEST SCORE. The combination of the two scores will give you an ENGLISH/WRITING SCALE SCORE, from 1 to 36, that will be used to determine your COMPOSITE SCORE mentioned earlier.

Using the English/Writing Scoring Table, find your ENGLISH SCALE SCORE on the left or right hand side of the table and your WRITING TEST SCORE on the top of the table. Follow your ENGLISH SCALE SCORE over and your WRITING TEST SCORE down until the two columns meet at a number. This number is your ENGLISH/WRITING SCALE SCORE and will be used to determine your COMPOSITE SCORE.

Step 3 Determine your ENGLISH/WRITING SCALE SCORE using the English/Writing Scoring Table on the following page:

English _____

Writing _____

English/Writing _____

ENGLISH/WRITING SCORING TABLE

ENGLISH SCALE SCORE	WRITING TEST SCORE											ENGLISH SCALE SCORE
	2	3	4	5	6	7	8	9	10	11	12	
36	26	27	28	29	30	31	32	33	34	32	36	**36**
35	26	27	28	29	30	31	31	32	33	34	35	**35**
34	25	26	27	28	29	30	31	32	33	34	35	**34**
33	24	25	26	27	28	29	30	31	32	33	34	**33**
32	24	25	25	26	27	28	29	30	31	32	33	**32**
31	23	24	25	26	27	28	29	30	30	31	32	**31**
30	22	23	24	25	26	27	28	29	30	31	32	**30**
29	21	22	23	24	25	26	27	28	29	30	31	**29**
28	21	22	23	24	24	25	26	27	28	29	30	**28**
27	20	21	22	23	24	25	26	27	28	28	29	**27**
26	19	20	21	22	23	24	25	26	27	28	29	**26**
25	18	19	20	21	22	23	24	25	26	27	28	**25**
24	18	19	20	21	22	23	23	24	25	26	27	**24**
23	17	18	19	20	21	22	23	24	25	26	27	**23**
22	16	17	18	19	20	21	22	23	24	25	26	**22**
21	16	17	17	18	19	20	21	22	23	24	25	**21**
20	15	16	17	18	19	20	21	21	22	23	24	**20**
19	14	15	16	17	18	19	20	21	22	23	24	**19**
18	13	14	15	16	17	18	19	20	21	22	23	**18**
17	13	14	15	16	16	17	18	19	20	21	22	**17**
16	12	13	14	15	16	17	18	19	20	20	21	**16**
15	11	12	13	14	15	16	17	18	19	20	21	**15**
14	10	11	12	13	14	15	16	17	18	19	20	**14**
13	10	11	12	13	14	14	15	16	17	18	19	**13**
12	9	10	11	12	13	14	15	16	17	18	19	**12**
11	8	9	10	11	12	13	14	15	16	17	18	**11**
10	8	9	9	10	11	12	13	14	15	16	17	**10**
9	7	8	9	10	11	12	13	13	14	15	16	**9**
8	6	7	8	9	10	11	12	13	14	15	16	**8**
7	5	6	7	8	9	10	11	12	13	14	15	**7**
6	5	6	7	7	8	9	10	11	12	13	14	**6**
5	4	5	6	7	8	9	10	11	12	12	13	**5**
4	3	4	5	6	7	8	9	10	11	12	13	**4**
3	2	3	4	5	6	7	8	9	10	11	12	**3**
2	2	3	4	5	6	6	7	8	9	10	11	**2**
1	1	2	3	4	5	6	7	8	9	10	11	**1**

Step 4 Determine your COMPOSITE SCORE by finding the sum of all your SCALE SCORES for each of the four sections: English only (if you do not choose to take the optional Writing Test) *or* English/Writing (if you choose to take the optional Writing Test), Math, Reading, and Science Reasoning, and divide by 4 to find the average. Round your COMPOSITE SCORE according to normal rules. For example, $31.2 \approx 31$ and $31.5 \approx 32$.

_____	+	_____	+	_____	+	_____	=	_____
ENGLISH *OR* ENGLISH/WRITING SCALE SCORE		MATHEMATICS SCALE SCORE		READING SCALE SCORE		SCIENCE SCALE SCORE		SCALE SCORE TOTAL

_____	÷ 4 =	_____
SCALE SCORE TOTAL		COMPOSITE SCORE

ANSWERS AND EXPLANATIONS

English Test Explanations

PASSAGE I

1. **The best answer is C.** The modifier *that* should be used to reference a specific year; because the actual year is not stated in the passage, the underlined portion should be changed to *this*, which clarifies the sentence and provides context. It is not appropriate here to use the article *a* before the noun *year*, so eliminate answer choice B. Omitting the underlined portion creates an incomplete sentence, so eliminate answer choice D.

2. **The best answer is F.** The adjective *careful* correctly modifies the verb phrase *monitoring of*. This choice also indicates the writer's direct involvement—answer choices G and H are not controlled by the writer, and answer choice J does not indicate any action on the part of the writer.

3. **The best answer is D.** It is correct to use the past tense verb *instigated* to match the tense used throughout the paragraph.

4. **The best answer is F.** This sentence is correct as written. It is in the active voice and correctly uses the infinitive *to witness*. Answer choice G suggests that witnessing the transformation was the method by which the author arrived home, which doesn't make sense. Answer choices H and J are awkward and do not effectively convey the intended idea.

5. **The best answer is D.** The phrase *upon our return* is redundant because the writer has already stated that they had arrived back home. Answer choices B and C are also redundant for the same reason.

6. **The best answer is H.** The phrase *the local deer population* correctly identifies the subject of the clause. The pronoun *they* is an unnecessary addition in both answer choices F and G. In answer choice J, *they* would have referred logically to the plants, which doesn't make sense.

7. **The best answer is A.** The focus of the paragraph is the condition of the vegetable garden. The writer is painting a detailed picture of the garden, so the level of detail is appropriate. The construction of the garden fence comes later and is irrelevant at this point, which eliminates answer choices B and C.

8. **The best answer is J.** The writer's use of the word *comical* conveys the idea of humor. Only answer choice J adds to that idea without being redundant, while also being grammatically correct.

9. **The best answer is B.** The phrase *needed to be remedied immediately* is a nonrestrictive clause. This means it could be eliminated from the sentence and the sentence would still make sense. Nonrestrictive clauses take the pronoun *which*. Answer choice C is wrong because *deterring* modifies *qualities* and therefore should not be separated from it by a comma.

10. **The best answer is F.** Answer choice F implies sophistication and detailed preparation, both of which indicate time-consuming planning. Answer choices G, H, and J are vague. They add very little to communicate the writer's message.

11. **The best answer is C.** Answer choices A, B, and D all indicate a sequence of time where the writer first goes to the hardware store and then arranges her tools, which she logically implies were purchased at the store. Answer choice C reverses the sequence and, therefore, doesn't make sense.

12. **The best answer is J.** Answer choices F, G, and H convey the same idea of a difficult task. While some are wordier than absolutely necessary, all three are grammatically correct. Answer choice J, however, is neither grammatically correct, nor idiomatic.

13. **The best answer is B.** Only answer choice B matches the tone of the rest of the passage, while providing the same degree of description. Answer choice D is too wordy and awkward, while answer choices A and C do not provide the level of detail offered in the rest of the passage.

14. **The best answer is G.** The descriptive clause *swift and beautiful leapers* should refer to the subject of the sentence. Only answer choice G has the subject *deer*. In the other choices, the subject is *the newness and appearance of the enclosure*, which cannot be modified by the clause *swift and beautiful leapers*.

15. **The best answer is A.** This question really asks which preposition should follow *we sit down*. A person sits down *to* eat something—in this case, a variety of vegetables. Simplifying the sentence allows you to hear the correct idiomatic form.

PASSAGE II

16. **The best answer is J.** The idea of surprise is already expressed by the word *unexpected*. Therefore, *surprisingly* is redundant and should be omitted.

17. **The best answer is B.** The adverb *truly* is used here as an interjection to express wonder. Answer choice B changes the tone to something more objectively descriptive. Because of this change, the choice is NOT acceptable.

18. **The best answer is F.** The clause that begins *as they enjoy* is not a complete sentence. Therefore, answer choices G and J should be eliminated. Answer choice H places a comma after *as*, which breaks up its clause. However, the phrase *Viewing these birds can be difficult* is a complete thought and can be separated from the following clause by a comma.

19. **The best answer is D.** The beginning of this paragraph focuses on a physical description of the Wood Thrush and where it likes to perch. Without the information about migration, a reader might assume these details explain when the bird is seen in the United States. This would be misleading, because the real explanation is the bird's migration pattern. Therefore, D is the correct answer choice.

20. **The best answer is G.** The singular possessive pronoun refers back to a singular subject: *the Wood Thrush*. Therefore, answer choices H and J are incorrect. The word *it's* is a contraction of *it is*, so answer choice F is incorrect.

21. **The best answer is B.** This introductory clause needs a preposition to indicate duration of time. Answer choices A, C, and D all have prepositions that place the action of nesting in a temporal sequence. Only answer choice B lacks the preposition, which makes the clause awkward and incomplete.

22. **The best answer is F.** The underlined portion is a complete thought. This means it does not require commas to separate its ideas. Answer choice H places too much emphasis on *in distinct parts*, which is not supported by the focus of the sentence.

23. **The best answer is C.** The writer wants to give the idea of the listener learning new things about the Wood Thrush's song *over time*. All of the answer choices have a sense of duration *except* answer choice C. Answer choice C implies that the listener learns something at the beginning of the song only; the learning process does not continue as the song continues. Therefore, answer choice C is the correct answer choice.

24. **The best answer is H**. The underlined portion introduces a subordinate clause that explains the action of the main clause (i.e. the bird has stopped singing because it's exhausted). All of the answer choices could be used in a subordinate clause, but, apart from answer choice H, they would need additional words to make grammatical sense. For example, answer choice J would have to read: *and so indicates* in order to make sense. Answer choice H contains a participle clause that stands on its own. The addition of the relative pronoun *that* (answer choice F) is unnecessary.

25. **The best answer is C.** The added sentence gives more detail about the songs that the writer begins to describe in sentences 4 through 6. Therefore, answer choices A and B are too early in the paragraph to be correct. By sentence 7, the topic has shifted to why the songs have ended. Only sentence 6 (answer choice C) places the additional information in the correct context with other details about the Wood Thrush songs.

26. **The best answer is F.** In the next paragraph, the writer describes the bird's continuing urge to sing, even after each act exhausts him. Only answer choice F introduces that idea to the reader. The other answer choices are either irrelevant, like answer choice H, or mis-focused, like answer choices G and J.

27. **The best answer is D.** The underlined portion is in the main clause of the sentence and needs an active verb. Because the writer is using the present tense, only answer choice D, *will start*, is consistent. Answer choices B and C are forms of the past tense. Answer choice A is a participle and would only be appropriate in a subordinate clause.

28. **The best answer is F.** The phrase *harmonious musical arrangement* is the most specific and precise because it gives the most detail about the sound itself. The other answer choices are vague and less descriptive.

29. **The best answer is D.** The *chosen female and subsequent offspring* belong to the Wood Thrush male. Therefore, the possessive form needs to be used. Answer choices B and C are incorrect because *chosen* is an adjective modifying *female*, indicating that the female is the Thrush's mate; the verb phrase, *are begging*, comes later.

30. **The best answer is G.** The subordinate clause describes what the female and offspring are begging the male to do. Consequently, the infinitive clause *to keep singing* is correct. Answer choice H implies that the female and offspring are the ones

singing. Answer choices F and J are awkward and grammatically incorrect.

PASSAGE III

31. **The best answer is C.** The singular noun *America* must be in the possessive form.

32. **The best answer is G.** The underlined portion contains the sentence's main verb. Answer choices F, H, and J all have active voice verbs that would be acceptable in the passage. Answer choice G has only an *–ing* participle, *being*, which doesn't work as a main verb, and is NOT acceptable.

33. **The best answer is D.** The writer explains that other members of the group wrote the journals. Therefore, no modifier is necessary before the word *journals*, and answer choice D is correct.

34. **The best answer is J.** The addition is not necessary because it is off topic. Both the preceding paragraph and the following paragraph focus on Sacajawea's life and personality. Clark's role may be true and interesting, but it doesn't really clarify Sacajawea's role. Instead, the addition distracts the reader from Sacajawea's contribution and is unnecessary.

35. **The best answer is B.** The original sentence contains two complete thoughts with two main verbs. Connecting these two clauses with a comma is called a comma splice and is grammatically incorrect. Answer choice A creates two separate sentences, which is acceptable. Answer choices C and D turn the second half of the sentence into a subordinate clause, which can be separated from a main clause by a comma.

36. **The best answer is J.** The most logical arrangement would place the two verbs close together so that the reader could easily follow the action. Therefore, answer choices G, H, and J are all preferable to F. Adjectives should also be placed close to the noun they modify. Therefore, answer choice J is better than G and H because *westward* is kept closest to *trek* in answer choice J. While H does keep the two verbs the closest, its construction is too awkward to be the best answer choice.

37. **The best answer is B.** *Its* is a possessive pronoun modifying *roster*, but refers back to the singular noun *expedition*. *It's* is a contraction of the subject/verb *it is*, which makes no sense here. Answer choice D is incorrect because it includes an extraneous comma and *their* is plural. Because it is a singular pronoun, *its'* does not exist in correct English.

38. **The best answer is J.** The simple past tense *was reunited* is consistent with the tense used in the rest of the passage.

39. **The best answer is D.** The interjections in answer choices A, B, and C do not add necessary information to the passage. More importantly, they are too casual and do not match the academic tone of the rest of the passage.

40. **The best answer is F.** Only answer choice F is grammatically correct. Answer choices G and H are wrong because they create incomplete sentences. Answer choice J places an unnecessary comma after *in addition to*, which makes it grammatically incorrect.

41. **The best answer is A.** The best answer choice is A because the final clause directly explains why her successful negotiations were important. While the whole sentence is consistent with the first paragraph, the final clause does not provide a direct link, so answer choice B is not correct. The clause is not humorous (answer choice C) or a lengthy description of Sacajawea's family (answer choice D).

42. **The best answer is J.** Because *1812* is directly followed by the phrase *as the year she died*, answer choices F and H are redundant. Answer choice G does not make sense.

43. **The best answer is C.** The sentence has to specify what the books and instruments were falling into, so answer choice D is incorrect. The pronouns in answer choices A and B are unclear referents. Only answer choice C correctly explains where the books and instruments were going.

44. **The best answer is F.** Only answer choice F accurately summarizes the main idea of the passage. Answer choice G is incorrect because it focuses on a detail, and it also contradicts the passage, which claims Sacajawea's lifespan is controversial. Answer choice H also focuses on a detail: Sacajawea's role as mother. While the passage does mention Sacajawea's calm manner several times (answer choice J), her demeanor is only important because it enabled her to be an important influence on American history.

45. **The best answer is D.** Answer choice D is correct because it places the information about the end of Sacajawea's life at the end of the passage. This way, the passage presents its information in a consistently chronological fashion. It also leads nicely into the last paragraph, which refers to the controversy introduced in Paragraph 4.

PASSAGE IV

46. The best answer is G. To take something *by storm* is an idiomatic expression meaning to overcome suddenly and with great force. None of the other answer choices make sense in this context.

47. The best answer is B. The underlined portion contains the main verb of the sentence. Therefore, answer choice D is incorrect because it contains a relative pronoun, which would make the phrase a relative clause. Answer choices A and C are vague – the reader does not know who is doing the needing. Answer choice B, however, contains a pronoun that means "anyone." This makes perfect sense in the context of the sentence.

48. The best answer is F. The sentence in question is important because it explains how to play the game of Sudoku, and Sudoku is the subject of the passage. Answer choice G is incorrect because it does not provide a solution to any particular game. Answer choices H and J are incorrect because the sentence is both relevant and to the point.

49. The best answer is C. In this sentence, *the solver's task* is the subject and *is* is the main verb. It is unnecessary to separate them with any punctuation.

50. The best answer is F. The subject here is *a level*, which is singular. Therefore, answer choices G and H, which use plural verbs, are incorrect. Answer choice F is also correct because it uses the present indicative *is*.

51. The best answer is D. In this sentence, *become* is the main verb and is modified by the adverb *eventually.* Answer choices A and B use the incorrect form of the adverb. Answer choice C turns *become* into the gerund *becoming*, which is incorrect for a main verb in a main clause.

52. The best answer is G. Here the underlined word serves as a connector between this sentence and the preceding sentence. This sentence gives specific details refining the ideas of the previous one. Therefore, answer choice G is best. Answer choice F suggests contrast that is inappropriate in this context. Answer choice H suggests consequence. Answer choice J is too vague and doesn't make sense here.

53. The best answer is C. The subordinate clause explains why the puzzle is aptly named. Answer choices A, B, and D all contain the idea of explanation, and therefore would be appropriate alternatives. Answer choice C suggests a contrast, which doesn't make sense in this context.

54. The best answer is G. The underlined verb needs to be in the present tense to be consistent with the rest of the passage. Answer choice J contains a mistaken form of *should have*, which is never correct English.

55. The best answer is A. The modifier *which* correctly specifies that the writer is referring to a unique, but unspecified, number. *That* is inappropriate because it would indicate that the writer had a specific number in mind, so eliminate answer choice B. The article *a* is too vague, and doesn't give any direction to the reader, so eliminate answer choice C. Omitting the modifier creates an awkward sentence, so eliminate answer choice D.

56. The best answer is F. This choice introduces the idea of a hypothetical mistake that the reader can then correct. This makes logical sense within the scope of the paragraph, which focuses on making mistakes while playing Sudoku. Answer choice G does not give a reason for the correction; therefore, it's unclear and should be eliminated. The same is true of answer choice J. Answer choice H introduces a time constraint that is not supported by the passage.

57. The best answer is A. Introductory dependent clauses must be separated from the main clause by a comma. Therefore, answer choice D should be eliminated. The preposition and participle (*By attacking*) also make the conjunctions *and* and *so* unnecessary. Therefore, answer choices B and C should be eliminated.

58. The best answer is H. The underlined word *sometimes* introduces the idea of irregularity. Euler was not occasionally Swiss nor did he invent Sudoku only on certain days, so answer choices F and J can be eliminated. On the other hand, it makes logical sense that only some people credit Euler with inventing the game. Therefore, answer choice H is the best answer.

59. The best answer is C. The beginning of this paragraph focuses on Euler's game and its relationship to Sudoku. However, with Sentence 4, the writer shifts focus to just Sudoku and the ways it challenges players. Therefore, answer choice C is the best answer because it recognizes this shift in focus.

60. The best answer is H. The beginning sentence of this paragraph ties the topic of the passage back

to the introductory paragraph. Therefore, answer choice H is best. The writer does not make explicit comparisons between Sudoku and crossword puzzles, so answer choice G can be eliminated. The writer also repeatedly encourages readers to try Sudoku, not to avoid it, so answer choice J can be eliminated. Answer choice F should be eliminated because the writer only hints at where Sudoku puzzles are located in a newspaper.

PASSAGE V

61. The best answer is C. In this example, *baby* is modifying *cries* and should be in the possessive.

62. The best answer is J. In this sentence, the two clauses contain two grammatically complete ideas. They can be made into two separate sentences as in answer choices F and G, or they can be connected by a conjunction as in answer choice H. Answer choice J, however, connects two complete sentences with a comma. This is called a "comma splice" and it is grammatically incorrect.

63. The best answer is B. This question tests parallel construction. Because *anxiety* is a noun, the idea that is linked to it should also be a noun in order to stay consistent. Therefore, answer choice B *confusion* is the best answer. The other answer choices are verbs.

64. The best answer is H. This choice clearly indicates what is being played back in slow motion (the film). Good writing is clear and not ambiguous. The remaining answer choices are awkward and unclear.

65. The best answer is D. A list of only two items, in this case *happiness and comfort*, does not need to be separated by commas. Therefore, answer choice D is the best answer.

66. The best answer is G. In this example, *sound* refers back to the plural noun phrase *new utterances*, so answer choice F can be eliminated. Because *sound* is the main verb in a relative clause, answer choice H should be eliminated. Finally, to stay consistent with the rest of the passage, *sound* needs to be in the present tense, so answer choice H can be eliminated.

67. The best answer is D. In this sentence, the two clauses contain two grammatically complete ideas. They can be made into two separate sentences joined by a semicolon as in answer choice B, or they can be connected by a conjunction as in answer choices A and C. Answer choice D is grammatically unacceptable.

68. The best answer is F. This sentence contrasts the results of failure to communicate with the results of success in the previous sentence. Therefore, answer choice F is the best answer because it is the only answer choice that indicates contrast. The other answer choices indicate causation (answer choices G and H) or comparison (answer choice J).

69. The best answer is D. The adjective *emerging* tells the reader that the child is just learning to talk. Therefore, any additional statement of that idea is redundant. Answer choice D is the best answer because the underlined passage is unnecessary.

70. The best answer is F. This is the only answer choice that lists specific needs. The other answer choices are not detailed and should be eliminated.

71. The best answer is A. The phrase explains the causal relationship between sign language and verbal language development. It does not provide any specific detail about individual signs, so answer choice B can be eliminated. The phrase does not contradict other information in the passage, so answer choice C can be eliminated. Finally, the phrase is already at the end of the paragraph, so answer choice D can be eliminated.

72. The best answer is G. The underlined word *distinct* is an adjective; it can't modify a verb, therefore answer choice J is incorrect. Also, in English, adjectives almost always come before the noun they modify. Since *distinct* correctly modifies *words*, answer choice G is the best answer. Answer choices F and H don't make sense.

73. The best answer is C. The verb phrase *may be able* must be followed by an infinitive verb, so answer choice C is the best answer. The other answer choices don't make grammatical sense.

74. The best answer is G. The section links the child's ability to follow commands to her own language acquisition. Answer choice F should be eliminated because the author doesn't say commands are the *only* way children learn language—in fact, the section on sign language would contradict this idea. Answer choice H is also contradicted by the passage and should be eliminated. Answer choice J should be eliminated because baby names are not mentioned at all.

75. The best answer is B. This sentence stands in contrast to the sentence before it by correcting the mistaken idea of *most people*. The underlined phrase should introduce that concept of contrast and correction; therefore, answer choice A should be eliminated. Answer choice C has the idea of contrast but implies that the two ideas are equally correct, so it should be eliminated. Answer choice D is similar to answer choice C. Only answer choice B clarifies the logical relationship between the two sentences, and is the best answer.

Mathematics Test Explanations

1. **The correct answer is D.** To solve this problem, first perform the operations within the absolute value signs, and then perform the subtraction, as follows:

$$|9-5|-|5-9|$$
$$=|4|-|-4|$$
$$=4-4=0$$

2. **The correct answer is H.** To solve this problem, first subtract the flat fee from the total bill:

$$190-25=165$$

The editor billed \$165 for his hourly work. Now, simply divide 165 by 30 to determine the number of hours he worked:

$$165 \div 30 = 5.5, \text{ or } 5\frac{1}{2}$$

3. **The correct answer is A.** The first step in solving this problem is to calculate the time it takes each runner to run 15 miles. Use the Distance = Rate × Time formula, as follows (because you are calculating the time, use the variable x):

A (5 mph)	B (5 mph)
$D=(r)(t)$ $15=5x$ $x=3$	$D=(r)(t)$ $15=6x$ $x=2.5$

Runner A takes 3 hours and Runner B takes 2.5 hours to run 15 miles. Therefore, it takes Runner A half an hour ($3-2.5=0.5$) longer to run 15 miles.

4. **The correct answer is H.** To solve this problem, add or subtract like terms, as follows:

$$x^2+60x+54-59x-82x^2$$
$$=(-82x^2+x^2)+(60x-59x)+54$$
$$=-81x^2+x+54$$

5. **The correct answer is D.** The perimeter is the distance around an object. You are given that the figure is composed of a square, which has congruent sides, and an equilateral triangle, which has congruent sides. You are also given that the length of one side of the square is 18 cm, which means that the length of each of the remaining sides is also 18. Therefore, the perimeter of $ABCDE$ is $18 \times 5 = 90$.

6. **The correct answer is K.** To solve this problem, use the FOIL method, as follows:

$$(6n-5)(n+4)$$
$$\text{First terms: } 6n^2$$
$$\text{Outside terms: } 24n$$
$$\text{Inside terms: } -5n$$
$$\text{Last terms: } -20$$

Now, put the like terms in order and simplify:

$$6n^2+24n-5n-20$$
$$=6n^2+19n-20$$

7. **The correct answer is C.** A quick way to solve this problem is to multiply Blair's current salary by 1.03, which will yield her new salary with a 3% raise (remember, the decimal equivalent of 3% is .03):

$$42,000 \times 1.03 = 43,260$$

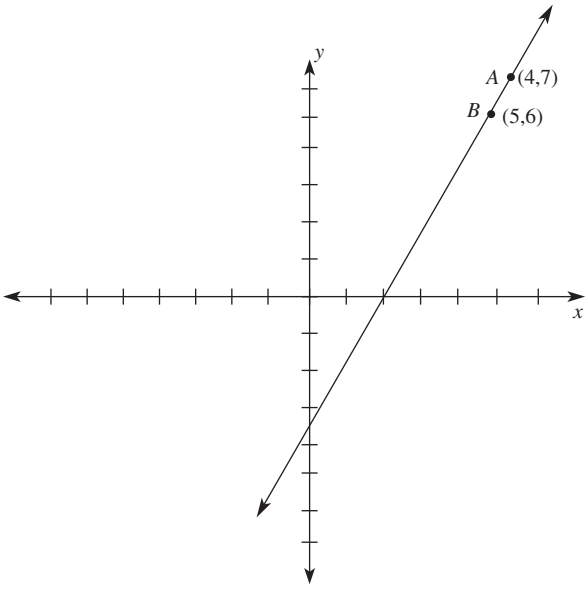
You could also multiply Blair's current salary by .03, then add the result to her current salary:

$$42,000 \times .03 = 1,260$$
$$42,000 + 1,260 = 43,260$$

8. **The correct answer is J.** One way to solve this problem is to simply set up the equation and solve for n, as follows:

$$(n-2)+(n-1)+n+(n+1)$$
$$+(n+2)+(n+3)=513$$
$$6n-2+(-1)+1+2+3=513$$
$$6n-3+6=513$$
$$6n+3=513$$
$$6n=510$$
$$n=85$$

9. The correct answer is C. One way to solve this problem is to sketch the (x, y) coordinate plane and plot the given points, as shown below:



A quick glance at the figure and the answer choices should indicate that point C would likely have coordinates (4,5) because none of the other options match the figure. Use the midpoint formula to calculate the coordinates, as follows:

$$5 = \frac{(6+x)}{2} \qquad 6 = \frac{(7+y)}{2}$$

$$10 = 6 + x \qquad 12 = 7 + y$$

$$4 = x \qquad 5 = y$$

10. The correct answer is J. Since the figure is a rectangle, the adjacent sides are perpendicular. Perpendicular lines have slopes that are negative reciprocals of each other, meaning that the product of their slopes is -1. Since there are four lines and four perpendicular angles, the product of the slopes is $(-1) \times (-1) \times (-1) \times (-1) = 1$.

11. The correct answer is D. You are given that Tom traveled 45 miles in 12 hours and that Jim traveled four times as far in one-third the time. That means that Jim traveled 45×4, or 180 miles in $12 \times (\frac{1}{3})$, or 4 hours. Therefore, Jim traveled $\frac{180}{4}$, or 45 miles per hour.

12. The correct answer is J. If you recognized that a, b, and c are supplementary to the interior angles of the triangle, you could have quickly arrived at the correct answer: $a + c = 180$; $a + b = 180$; $b + c = 180$; therefore, $a + b + c = 360$. You could also have calculated the sum of a, b, and c by first finding their values. Because 2 of the interior angles total 124 (43 + 81), the third angle must measure $180 - 124 = 56$. Therefore, angle c must measure $180 - 56 = 124$, angle a must measure $180 - 43 = 137$, and angle b must measure $180 - 81 = 99$. The sum is $137 + 99 + 124 = 360$.

13. The correct answer is E. According to the information given in the table, 30 students voted for the name Spartans. Therefore, 30 out of 200 students voted for Spartans. Because you are looking for a percentage, set up a proportion and solve for x, as follows:

$$\frac{\text{30 students}}{\text{200 students}} = \frac{x \text{ percent}}{\text{100 percent}}$$

$$3,000 = 200x$$

$$15 = x$$

14. The correct answer is H. According to the information given in the table, 40 students voted for the name Lions. Therefore, 40 out of 200 students voted for Lions. First, calculate the percentage of the whole pie chart represented by Lions by setting up a proportion and solving for x, as follows:

$$\frac{\text{40 students}}{\text{200 students}} = \frac{x \text{ percent}}{\text{100 percent}}$$

$$4,000 = 200x$$

$$20 = x$$

You now know that 20% of the students voted for Lions. Because there are 360° in a circle, find 20% of 360:

$$360 \times 0.2 = 72$$

15. The correct answer is D. According to the information given in the table, 50 students voted for the name Knights. Therefore, 50 out of 200 students polled voted for Knights. First, calculate the percentage of those students polled who voted for Knights by setting up a proportion and solving for x, as follows:

$$\frac{\text{50 students}}{\text{200 students}} = \frac{x \text{ percent}}{\text{100 percent}}$$

$$5,000 = 200x$$

$$25 = x$$

You now know that 25% of the students polled voted for Lions. You are given that there are 3,000 students at Center High School, so find 25% of 3,000:

$$3000 \times 0.25 = 750$$

16. The correct answer is J. To solve this problem, calculate the area of the sides:
Top/Bottom:

$$4 \times 4 = 16$$

$$2(16) = 32$$

Sides:

$$3 \times 4 = 12$$

$$4(12) = 48$$

$32 + 48 = 80$, the total surface area.

17. The correct answer is C. In the equation of a line, $y = mx + b$, m is the slope. Recall that parallel lines have the same slope. Therefore, because the slope of the given line is $\frac{2}{5}$, the slope of any line parallel to that line will also have a slope of $\frac{2}{5}$.

18. The correct answer is J. To solve this problem, use the formula for the area of a circle: $A = \pi r^2$. Substitute 2.5 for the radius and solve (remember that $\pi \cong 3.14$):

$$r = 2.5$$

$$A = \pi(2.5)^2$$

$$A = \pi(6.25)$$

$$A \cong 19.625 \cong 19.63$$

19. The correct answer is B. To quickly solve this problem, recall the perfect squares that are closest to $\sqrt{42}$:

$$\sqrt{36} = 6$$

$$\sqrt{49} = 7$$

Because the question asks for the largest integer *less* than $\sqrt{42}$, the correct answer must be 6.

20. The correct answer is G. The first step in solving this problem is to calculate the total surface area of Amanda's room. Each wall measures 12 by 14, so each wall has an area of $12 \times 14 = 168$. There are 4 walls, so the total area is $168 \times 4 = 672$. Now, subtract the area that will not be painted: $3 \times 4 = 12$ for the window, and $3 \times 7 = 21$ for the door; $168 - 12 - 21 = 639$. Because Amanda needs to cover 639 square feet, and each can of paint covers 350 to 400 square feet, she will need a minimum of 2 cans of paint.

21. The correct answer is A. To answer this question, remember that when you divide exponential values with the same base you must subtract the exponents. Simplify the expression as follows:

$$\frac{3x^3}{3x^9}$$
$$= \frac{x^3}{x^9}$$
$$= x^{-6}$$

22. The correct answer is K. To solve this problem, factor the quadratic equation as follows:

$$a^2 + 2a - 8 = 0$$
$$(a + 4)(a - 2) = 0$$
$$a + 4 = 0 \quad a - 2 = 0$$
$$a = -4 \quad a = 2$$

The values of a are -4 and 2.

23. The correct answer is C. Calculate the slope by finding the change in y-values over the change in x-values using the coordinates for points A and C:

$$\frac{4 - (-4)}{4 - (-4)} = \frac{8}{8} = 1$$

24. The correct answer is K. You are given that the fixed costs are \$900, which means that the cost of printing will be \$900 *plus* the cost of printing each textbook. The cost of printing each textbook is determined by multiplying the number of textbooks, t, by the cost per textbook, \$1.50. Therefore, the cost of printing each textbook is $1.50t$, and the cost of printing t textbooks in one week is $900 + 1.50t$.

25. The correct answer is D. A right triangle in which the length of the longer leg is $\sqrt{3}$ times the length of the shorter leg is a 30°–60°–90° right triangle. Another property of this type of right triangle is that the hypotenuse is 2 times the length of the shorter leg. So, this right triangle has lengths x, $x\sqrt{3}$, and $2x$. Its perimeter is $x + x\sqrt{3} + 2x = 15$, or $3x + x\sqrt{3} = 15 + 5\sqrt{3}$. Subtract $5\sqrt{3}$ from both sides to get $3x = 15$; $x = 5$.

26. The correct answer is G. To solve this problem, note that the numerators are equivalent. Because

this is an equation where the fractions are equal to one another, the denominators must also be equivalent. Set up an equation for the denominators and solve for y as follows:

$$y\sqrt{11} = \sqrt{11}$$
$$y = 11/\sqrt{11}$$
$$y^2 = (\frac{11}{\sqrt{11}})^2$$
$$y^2 = \frac{121}{11} = 11$$
$$y = \sqrt{11}$$

You could also have quickly arrived at the correct answer by noticing that $\sqrt{11} \times \sqrt{11} = 11$, so y must equal $\sqrt{11}$.

27. The correct answer is A. To solve this problem, first determine the size of each bucket. Because the total capacity of 12 buckets is g gallons, each bucket can hold $(\frac{1}{12})g$, or $\frac{g}{12}$ gallons. Because the total capacity of 8 buckets is g gallons, then each of those buckets can hold $(\frac{1}{8})g$, or $\frac{g}{8}$ gallons. If the total capacity of 4 buckets is g gallons, then each bucket is $(\frac{1}{4})g$, or $\frac{g}{4}$ gallons. Therefore, the capacity of the smallest buckets is $\frac{g}{12}$.

28. The correct answer is F. To solve this problem, remember that the formula for the circumference of a circle is C = $2\pi r$, and the formula for the area of a circle is $A = \pi r^2$, where r is the radius. You are given that the circumference is $.5\pi$. Calculate the value of r as follows:

$$2\pi r = 0.5\pi$$
$$r = 0.25$$

Now, substitute 0.25 into the formula for the area and solve:

$$A = \pi r^2$$
$$= \pi(0.25)^2$$
$$= 0.0625\pi$$

29. The correct answer is E. The volume of a cube is calculated by multiplying the length × width × height. You are given that Cube X has an edge length of 2. Because Cube Y has an edge length triple that of Cube X, the volume of Cube Y is $6 \times 6 \times 6$, or $6^3 = 216$ cubic inches.

30. The correct answer is G. To solve this problem, substitute the given values into the formula as follows:

$$A = 8,000(1 + 0.05)^3$$
$$= 8,000(1.157625)$$
$$= 9,261$$

31. The correct answer is D. To solve this problem, substitute the given values into the formula as follows:

$$S = 2\pi r^2 + 2\pi rh$$
$$= 2\pi(1)^2 + 2\pi(1)(3)$$
$$= 8\pi$$

32. The correct answer is F. To solve this problem, substitute the given values into the function as follows:

$$f(x) = 3x + 5 \quad \text{and} \quad g(x) = x^2 - x + 7$$

Therefore, $f(g(x)) = f(x^2 - x + 7)$. Substitute $x^2 - x + 7$ for x and solve:

$$f(g(x)) = 3(x^2 - x + 7) + 5$$
$$= 3x^2 - 3x + 21 + 5$$
$$= 3x^2 - 3x + 26$$

33. The correct answer is B. To solve this problem, first calculate the total number of touchdowns as follows:

2 games with 0 touchdowns = 0 touchdowns

3 games with 1 touchdown = 3 touchdowns

3 games with 2 touchdowns = 6 touchdowns

5 games with 3 touchdowns = 15 touchdowns

2 games with 4 touchdowns = 8 touchdowns

1 games with 5 touchdowns = 5 touchdowns

$0 + 3 + 6 + 15 + 8 + 5 = 37$ touchdowns

Now, calculate the average number of touchdowns per game. You are given that there were 16 games,

so the average is $37 \div 16 = 2.3125$, rounded to the nearest tenth is 2.3.

34. The correct answer is H. To solve this problem, first calculate the total measure of all the angles. The first triangle is a right triangle, which means that $a + b = 90°$. The sum of the interior angles of any triangle is 180°, so $c + d + e = 180°$. The total measure of the angles, then, is $90° + 180°$, or 270°. Calculate the average: $270° \div 5 = 54°$.

35. The correct answer is E. Because the entire value $4x^4$ is being raised to the 4th power, you must multiply the exponents: $4 \times 4 = 16$, so $(x4)^4 = x^{16}$. Therefore, you can eliminate answer choices A, B, and D. Next, you must raise 4 to the 4th power: $4^4 = 256$, and the correct answer is $256x^{16}$.

36. The correct answer is G. Inequalities can be solved like equations, by isolating the variable on one side. Remember, though, that when you multiply an inequality by a negative number, you must reverse the sign:

$$3x - 6 > 6x + 9$$
$$-3x > 15$$
$$x < -5$$

37. The correct answer is C. To find the point of intersection, set the equations equal to each other: $2x^2 + 6 = -4x^2 + m$. Add $4x^2$ to both sides: $6x^2 + 6 = m$. Since the x-coordinate of the point of intersection is $\sqrt{5}$, then $6(\sqrt{5})^2 + 6 = m$ and $m = 36$.

38. The correct answer is J. The first step in solving this problem is to calculate the length of the hypotenuse. If you recognize that this is a special triangle, you know that the hypotenuse is 5. You could also let the answer choices guide you, since they each include some combination of 3, 4, and 5. Otherwise, use the Pythagorean theorem, as follows:

$$a^2 + b^2 = c^2$$
$$3^2 + 4^2 = c^2$$
$$9 + 16 = c^2$$
$$25 = c^2$$
$$c = 5$$

Next, recall that cosine is equal to the length of the adjacent side over the length of the hypotenuse. Therefore, $\cos \angle Z$ is $\frac{3}{5}$.

39. The correct answer is C. The best way to solve this problem is to use the process of elimination. Remember that you are looking for the answer that is NOT true! First, determine the common difference of the sequence: $16+(-5) = 11$; $11+(-5) = 6$; $6+(-5) = 1$. The common difference is -5, so eliminate answer choice D. Now, consider each of the remaining answer choices:

Answer choice A: The common difference is -5, so the fifth term is $1 + (-5) = -4$; eliminate answer choice A.

Answer choice B: The sum of the first five terms is $16 + 11 + 6 + 1 + -4 = 30$; eliminate answer choice B.

Answer choice C: If the fifth term is -4, the sixth term is $-4+(-5) = -9$, and the seventh term is $-9+(-5) = -14$. Answer choice C is correct because it is NOT true.

40. The correct answer is G. The first step in solving this problem is to recognize that the average will be equivalent to the volume divided by the number of oxygen molecules. Set up the fraction and solve, keeping in mind that when you divide exponential values with the same base you subtract the exponents:

$$\frac{6 \times 10^{14}}{3 \times 10^7} = 2 \times 10^7$$

41. The correct answer is B. Cotangent is defined by the length of the adjacent side over the length of the opposite side. Therefore, the cotangent of $\angle B$ is $\frac{x}{z}$.

42. The correct answer is H. A rational number is any number that can be written as a ratio; in other words, a fraction. So simply determine the fraction that is exactly halfway between $\frac{1}{6}$ and $\frac{1}{2}$. One way to do this is to find the average, as follows:

$$\frac{\left(\frac{1}{2} + \frac{1}{6}\right)}{2} = \frac{2}{6} = \frac{1}{3}$$

Another way to solve this problem is to find the Lowest Common Denominator (LCD):

$$\frac{1}{2} = \frac{3}{6}, \text{so the LCD is 6.}$$

The value that is halfway between $\frac{1}{6}$ and $\frac{3}{6}$ is $\frac{2}{6}$, which is equivalent to $\frac{1}{3}$.

43. The correct answer is A. This problem tests your knowledge of absolute values and inequalities. The inequality $|6-2x|>5$ can be written as two separate inequalities: $6 - 2x > 9$ *or* $6 - 2x < -9$. Because the original inequality was a greater than, the word in between the two new inequalities must be "or." This means that $6 - 2x > 9 = -2x > 3 = x < -\frac{3}{2}$, *or* that $6 - 2x < -9 = -2x < -15 = x > \frac{15}{2}$. (Remember to reverse your inequality when multiplying or dividing by negative numbers.) The only answer choice that fits one of the inequalities is -2, which is less than $-\frac{3}{2}$. There are no answer choices that are greater than $\frac{15}{2}$.

44. The correct answer is J. A regular pentagon (5-sided figure) is a pentagon with equal sides and equal angles. If each side of the pentagon has a length of 12, its perimeter is 60. For the square to have a perimeter of 60, each side of the square must have a length of 15. If each side of the square has length of 15, its area is 225.

45. The correct answer is B. You are given that no table can be empty, which means that there must be at least one student seated at each table. There are 20 students in all and 10 tables, so if one student is seated at each table, you have accounted for 10 students. Now simply fill tables with the remaining 10 students until no students remain:

Table 1 already has 1 student; add 3 more so that it is filled. You now have 7 more students to seat.

Table 2 already has 1 student; add 3 more so that it is filled. You now have 3 more students to seat.

Table 3 already has 1 student; add the 3 remaining students so that it is filled.

You have accounted for all of the students and have filled 3 tables. You cannot fill more than 3 tables with students while leaving no tables empty.

46. The correct answer is K. One way to solve this problem is to pick numbers for x and y. You are given that $x < y$, so make $x = 2$ and $y = 3$. Now, substitute those values into the absolute value given: $|x - y| = |2 - 3| = |-1| = 1$.

Now, substitute the values into each answer choice. The answer choice that yields a value of 1 will be correct:

Answer choice F: $2 + 3 = 5$; eliminate answer choice F.

Answer choice G: $-(2 + 3) = -5$; eliminate answer choice G.

Answer choice H: $\sqrt{2} - 3 \neq 1$; eliminate answer choice H.

Answer choice J: $2-3 = -1$; eliminate answer choice J

Answer choice K: $-(2 - 3) = -(-1) = 1$. Answer choice K is correct. For any values of x and y when $x < y$, $|x - y| = -(x - y)$.

47. The correct answer is D. You can solve this problem by first calculating the area of the trapezoid. Use the formula $A = \frac{1}{2}(b_1 + b_2)h$:

$$A = \frac{1}{2}(4 + 8)3$$

$$A = \frac{1}{2}(12)3$$

$$A = \frac{1}{2}(36)$$

$$A = 18$$

Next, calculate the area of the rectangle: $3 \times 4 = 12$. Because the shaded regions comprise the difference between the area of the trapezoid and the area of the rectangle, the area of the shaded regions is $18 - 12 = 6$.

48. The correct answer is J. To solve this problem, find the missing side lengths. Because the lengths 2 and 4 make up the entire left vertical side of the figure, you can deduce that the entire right vertical side of the figure will be $2 + 4 = 6$. Likewise, because the entire bottom horizontal side of the figure is 8, you can deduce that the missing horizontal length will be 2, because $8 - 6 = 2$. Therefore, the perimeter of the figure is $6 + 6 + 8 + 8 = 28$.

49. The correct answer is D. Since c is directly proportional to s^2, then $c = as^2$, for some constant a. Given that $c = \frac{7}{16}$ when $s = \frac{1}{4}$, you can solve for a, as follows:

$$\frac{7}{16} = a\left(\frac{1}{4}\right)^2$$

$$\frac{7}{16} = \left(\frac{1}{16}\right)a$$

$$a = 7$$

You now know that $a = 7$, and $c = as^2$. Substitute 175 for c and 7 for a and solve for s, as follows:

$$175 = 7s^2$$
$$25 = s^2$$
$$s = 5$$

50. The correct answer is G. You must know something about the values of cosine functions at certain values of α to answer this question correctly. Work through each of the answer choices, as follows:

Answer choice F: ($\frac{2\pi}{3} \leq \alpha \leq \pi$) When α is $\frac{2\pi}{3}$, $\cos\alpha = -\frac{1}{2}$; when α is π, $\cos\alpha = -1$. The value −0.385 is not between $-\frac{1}{2}$ and −1, so eliminate answer choice F.

Answer choice G: ($\frac{\pi}{2} \leq \alpha \leq \frac{2\pi}{3}$) When α is $\frac{\pi}{2}$, $\cos\alpha = 0$; when $\alpha\frac{2\pi}{3}$, $\cos\alpha = -\frac{1}{2}$. The value −0.385 is between 0 and $-\frac{1}{2}$, so answer choice G is correct.

51. The correct answer is B. The first step in solving this problem is to determine how many integers between 10 and 99, inclusive, will have 0 as at least one digit, as follows:

10, 20, 30, 40, 50, 60, 70, 80, 90

There are 9 integers that will have 0 as at least one digit. Next, because there are a total of 90 numbers in the given range, the probability of choosing one of those 9 numbers is $\frac{9}{90}$, which simplifies to $\frac{1}{10}$.

52. The correct answer is J. The first step in solving this problem is to calculate the diameter of the circle, which corresponds to the diagonal of the rectangle. Use the Pythagorean Theorem to determine the length of the diagonal:

$$12^2 + 16^2 = c^2$$
$$144 + 256 = c^2$$
$$400 = c^2$$
$$20 = c$$

To find the area of the circle, you need the radius, which is equal to one-half the diameter, or 10. The formula for the area of a circle is $A = \pi r^2$, so the area is $\pi r^2 = 100\pi$.

53. The correct answer is E. By definition, $\log_a(xy)^2 = 2\log_a(xy) = 2(\log_a x + \log_a y)$. Because you are given that $\log_a x = n$ and $\log_a y = p$, the correct answer is $2(n + p)$.

54. The correct answer is J. You are given that a is a number with units digit x and tens digit y. Therefore, x is equivalent to 10 times y, and $a = xy = 10x + y$. You are given that b is formed by reversing the digits of a. Therefore, $b = yx = 10y + x$. Set up an equation and solve for $a - b$ as follows:

$$\begin{aligned} a - b &= (10x + y) - (10y + x) \\ &= 10x + y - 10y - x \\ &= 9x - 9y \\ &= 9(x - y) \end{aligned}$$

55. The correct answer is E. To solve this problem, substitute $a + b$ for a in $(a^2 - 2)$ as follows:

$$\begin{aligned} &(a^2 - 2) \\ &= (a + b)^2 - 2 \\ &= (a + b)(a + b) - 2 \\ &= a^2 + ab + ab + b^2 - 2 \\ &= a^2 + 2ab + b^2 - 2 \end{aligned}$$

56. The correct answer is J. To solve this problem, cross multiply and make the necessary substitution, as follows:

$$\begin{aligned} &\frac{1}{(1 + i)} \times \frac{(1 - i)}{(1 - i)} \\ &= \frac{1(1 - i)}{(1 + i)(1 - i)} \\ &= \frac{1 - i}{1 - i^2} \\ &= \frac{1 - i}{1 - (-1)} \\ &= \frac{1 - i}{2} \end{aligned}$$

57. The correct answer is A. To solve this problem, let t be Amy's time in 2005. Because her time decreased by 10% in 2006, her time in 2006 was $(.9)t$. Her time in 2007 would be $.8(.9t) = .72t$, which means that her time in 2007 was 72% of her time in 2005, so her time decreased by 28%.

58. The correct answer is K. According to the question stem, the sum of the sequence is given by

$\frac{x}{(1-y)}$. Therefore, sum $= \frac{x}{(1-y)}$. To solve this problem, first substitute the given values into the equation and solve for x:

$$200 = \frac{x}{(1-0.15)}$$

$$200 = \frac{x}{0.85}$$

$$170 = x$$

The first term of the sequence, x, is 170. To find the value of the second term, multiply 170 by the common ratio, 0.15: $170 \times 0.15 = 25.5$.

59. The correct answer is E. You are asked for integer values of a, which means that a must be a whole number. The first step will be to change $\frac{1}{11}$ to $\frac{2}{22}$ and $\frac{1}{8}$ to $\frac{2}{16}$; now it is easy to see that if $\frac{2}{a}$ is between $\frac{2}{22}$ and $\frac{2}{16}$, a can be equal to 21, 20, 19, 18, or 17.

60. The correct answer is G. In 3 fair coin tosses, there are exactly 8 equally likely outcomes: HHH, HHT, HTH, HTT, THH, THT, TTH, and TTT. Of these 8 outcomes, 3 include exactly 2 heads, so the probability of obtaining exactly 2 heads is $\frac{3}{8}$.

Reading Test Explanations

PASSAGE I

1. **The best answer is B.** The narrator begins by describing how silence was a concept unfamiliar to her. It made her uncomfortable: "I knew silence as a lack of something." The passage concludes by showing that her silence on the bus allowed Jaime to gather the courage to speak to the narrator, and that this experience had changed her long held opinion of silence.

2. **The best answer is F.** The tone of the passage indicates that the narrator is initially uncomfortable with silence in general. When the narrator sat next to Jaime, his quietness made her "feel a little awkward." This information best supports answer choice F.

3. **The best answer is D.** According to the passage, the narrator "never understood the value of silence." She considered silence "a lack of something," and goes on to say that, "maybe I was even a little afraid of the emptiness it created—the aural darkness where forgiveness never happened." These details best support answer choice D.

4. **The best answer is G.** The passage states that, "I knew silence as a lack of something." It goes on to say about silence that the narrator was "a little afraid of the emptiness it created." In addition, once the narrator encounters Jaime, he did not speak for several minutes. These details best support answer choice G.

5. **The best answer is B.** The narrator says that she didn't believe in silence as a healer before she met Jaime. However, the lines referenced in the question indicate a change of heart that is best expressed in answer choice B.

6. **The best answer is H.** According to the passage, "Buses have their own unique demographic: each crowd is unlike any other." This best supports answer choice H.

7. **The best answer is A.** According to the passage, "My university required a community service stint to graduate … I'd heard that the local YMCA was a good resource … I made my way to the YMCA looking for easy credits." These statements best support answer choice A.

8. **The best answer is J.** The passage states that, "his eyes were alert. And they were glued on me." These statements indicate that Jaime was alert, not sleepy or bored, so eliminate answer choices F and G. Because Jaime didn't speak to the narrator for several minutes, he could be described as shy, but he certainly could not be described as outgoing; eliminate answer choice H.

9. **The best answer is C.** The passage states that, "Within the parameters of my innocent world, I knew silence as a lack of something." This, along with other sentiments expressed by the author, suggests that the author's view of silence was limited or restricted. The other answer choices are not supported by the context.

10. **The best answer is F.** The passage states that, "On the bus, most everyone was either asleep or totally oblivious." This detail best supports answer choice F.

PASSAGE II

11. **The best answer is A.** As stated in the first paragraph of the passage, "The Republic had been under considerable stress for several years before Caius Julius was born, thus he did not create the fissures that led to the collapse of the Roman Republic, though he did capitalize on them brilliantly." This sentence indicates that Julius Caesar used conflicts of the past to enhance and promote his own power.

12. **The best answer is H.** The passage states that, "Caius Julius Caesar is popularly considered the founder of the Roman Empire, though it would be more accurate to consider his political rise as marking the end of the Roman Republic." From this statement, one can assume that because Caesar's political rise marked the end of the Roman Empire, Caesar exploited the political problems of the Roman Empire in order to gain power. The other answer choices do not reflect the main idea of the paragraph.

13. **The best answer is D.** The passage states, "Roman armies were traditionally made up of small landholders." This statement, along with the phrase in the question stem, indicates that unless a farmer owned a certain amount of land, he could not join the military, thereby establishing a close traditional relationship between farming and military service.

14. **The best answer is F.** As indicated by the passage, aristocratic fear of political reform was the reasoning behind the brothers' murders. Tiberius Gracchus was murdered over his determination to rid the Roman Empire of the military enrollment land requirement, while his brother, Gaius Gracchus, also caused trouble due to his rebellious ideas. "When the younger brother, Gaius Gracchus, began his reforms 10 years later, he was

able to extend political rights to the lower classes and reduce opportunities for bribery and corruption among the upper classes. He, too, was murdered for pushing the system too far." This information best supports answer choice F.

15. **The best answer is B.** As stated in the passage, "Marius also challenged the traditional structure of the army where nobility were regularly given authority over lower-class officers with more experience and ability." This indicates that Marius saw more value in ability than social class.

16. **The best answer is J.** The passage states, "Sulla was allied with the Roman Senate, who feared Marius' ambition and influence with the masses. Sulla believed in strengthening the power of the Senate against the popular Assembly. To this end, he marched his armies against Rome, defeating Marius and establishing himself as Dictator. While Sulla eventually resigned the dictatorship peacefully, he had exiled or killed thousands of political opponents during his reign." Answers F, G, and H are all found within the passage. Answer choice J is the only answer not found within the passage.

17. **The best answer is A.** The last paragraph of the passage opens by contrasting Julius Caesar's ties to aristocracy to his actions as a reformer. As indicated by the passages, Julius Caesar supported many ideas that were exactly the opposite of what his family ties desired. This contrast helps to illustrate that Julius Caesar was a true political reformer.

18. **The best answer is F.** Throughout the passage the author cites the actions of several different reformers, including Julius Caesar as well as the Gracchi brothers. Every reformer detailed by the author was assassinated. Because the author includes the stories and outcomes of these reformers, one can infer that many Roman reformers were eventually assassinated.

19. **The best answer is D.** The passage states, "Like the Gracchi brothers, Julius Caesar supported the redistribution of public lands to the poor and protected the grain supply (a large part of the unemployment dole). In his armies, he promoted ability before social rank. He was also widely seen to support the middle and lower classes against the privileges of the aristocracy, namely the Senate." Answer choices A, B, and C are all political reforms mentioned by the passage.

20. **The best answer is H.** The passage closes with the following statement in regard to the aftermath of Julius Caesar's assassination; "The resulting political chaos ultimately led to the rise of Augustus Caesar as emperor, effectively ending the Roman Republic forever." The other answer choices are not supported by the passage.

PASSAGE III

21. **The best answer is C.** The passage states in the last paragraph that *The Táin* is an epic of great value. "If the images meet a cultural need, they can come back to life as living artistic works. During the English occupation of Ireland up through the Irish Revolution, many artists plumbed the depths of Irish mythology to create what they saw as an image of Ireland free of English cultural repression. Writers still call on *The Táin* for inspiration, just as ancient Irish bards once called on the ghost of Fergus to tell them the true story of the Cattle Raid of Cooley." The passage makes it clear that the epic is of significant value to Irish history.

22. **The best answer is J.** The passage answers the questions presented in answer choices F, G, and H. The question regarding vernacular literature is answered with the following: "Ireland has the oldest vernacular literature in Europe. Where other early European authors wrote their literary works in Latin, the Irish began writing down their stories in their own language starting at least as early as the 6th century A.D. and continuing to the modern day." Vernacular language is defined by the passage as stories written down in one's native language. In regard to Queen Medb's desire to acquire a magic bull, her "insistence on equaling the property of her husband was, therefore, for reasons far more serious than vanity… Wives were considered legal equals to men if they came into their marriage with as much or more property than their husbands. If she had less, the wife would be a legal dependent of her husband and, like a child, would have limited rights of her own." Finally, the passage explains that *The Táin Bó Culainge* means "The Cattle Raid of Cooley." Answer choice J is the only question not answered by information in the passage.

23. **The best answer is A.** In regard to why the epic plays an important role in Irish literature, the passage states, "During the English occupation of Ireland up through the Irish Revolution, many artists plumbed the depths of Irish mythology to create what they saw as an image of Ireland free of English cultural repression." This information best supports answer choice A.

24. **The best answer is F.** The second paragraph serves to describe The Cattle Raid of Cooley. In its narrowest sense, the raid refers to a series of battles fought by the northern Irish province of Connacht to steal a magic bull from the neighboring province of Ulster. However, the cycle includes many other legends that together tell the national story of the people of Ulster, especially during the reign of the great Ulster king, Conchobor mac Nessa.

25. **The best answer is D.** Of the four answer choices, A, B, and C can all be found in the passage. "The *remscéla* tell how the bulls were originally two pig-keepers who knew magic; the stories explain how the Ulstermen came to be cursed with debilitating pain whenever their country was in danger. They explain who Cú Chulainn was and how he got his name. And, most poignantly, the *remscéla* tell how Fergus lost his crown and why he agreed to fight against his countrymen. This last story, told in the tragic legend of Deirdre and the Sons of Usnech, is one of the most striking of all the Irish myths." Answer choice D is the only choice not mentioned by the passage.

26. **The best answer is G.** As stated in the last paragraph of the passage, "Over twelve hundred years old, *The Táin* is certainly an epic work—but epic doesn't necessarily mean "dead." If the images meet a cultural need, they can come back to life as living artistic works … Writers still call on *The Táin* for inspiration, just as ancient Irish bards once called on the ghost of Fergus to tell them the true story of the Cattle Raid of Cooley." Not only does this segment of the passage establish answer choices F, H, and J as elements of *The Táin,* but it also states that an epic isn't dead, as long as it still has cultural significance.

27. **The best answer is D.** In an effort to explain why Queen Medb wanted a magic bull, the passage states the following: "Wives were considered legal equals to men if they came into their marriage with as much or more property than their husbands. If she had less, the wife would be a legal dependent of her husband and, like a child, would have limited rights of her own. Irish queens were used to having their own political autonomy and making their own political deals." This information best supports answer choice D.

28. **The best answer is J.** When speaking of love triangles, the passage gives the following as examples, "Think of the romances of King Arthur, Queen Guinevere, and Sir Lancelot, or those of Tristran, Isolde, and her husband, King Mark. In *The Táin,* Queen Medb takes up with the warrior Fergus, with the approval of her husband, in order to guarantee Fergus' allegiance during the war."

29. **The best answer is A.** As indicated by the passage, there are strong thematic similarities between the King Arthur story and *The Táin*. Specifically, the passage states, "Still, even Medb's daughter, Finnabair, is tied linguistically to the Arthurian legend: Finnabair and Guinevere are different spellings of the same name." This statement is used to further support the passages earlier tie between the two stories; "Thematically, the work explores several great issues that would occupy medieval authors for over six hundred years. One of the most important was the lovers' triangle between the king, the queen, and the warrior hero. Think of the romances of King Arthur, Queen Guinevere, and Sir Lancelot, or those of Tristran, Isolde, and her husband, King Mark." These statements best support answer choice A.

30. **The best answer is H.** The passage explains that Medb's dedication to acquiring one of the two magic bulls of Ulster was because not only did her husband have the other magic bull, but if she acquired a bull as well, she would attain the same level of political power as her husband. Otherwise, she would be at his mercy. Due to these circumstances, Medb was very ambitious in her pursuits. The other answer choices are not supported by the passage.

PASSAGE IV

31. **The best answer is D.** Of the four answer choices, choices A, B, and C can all be found in the first paragraph of the passage. This passage reads, "Homeopathy is a system for treating physical disease and other ailments using the theory of treating "like with like." In practice, homeopathic medicine seeks substances that mimic an ailment's symptoms; this sameness is considered "likeness." The substance is then diluted to infinitesimal amounts and administered to the patient in order to cure the problem. Homeopathic treatment is currently in use for everything from cancer to colds and flu, though many scientists remain heavily skeptical about its efficacy." Not only is answer choice D not mentioned in this paragraph, but the passage indicates that, "Modern scientists have been unable to find any evidence to support the theory of molecular memory," which is the opposite of the statement asserted by answer choice D.

32. The best answer is F. The passage states that, "Hahnemann had hypothesized that shaking the solution after each dilution would imprint the molecular "memory" of the original substance into the solution, which would allow the diluted dose to be effective without the possibility of overdose or adverse side effects." This information best supports answer choice F.

33. The best answer is C. As stated in the passage, "Hahnemann first stumbled upon his theory when he was investigating a common treatment for malaria, cinchona bark. Modern scientists now know that cinchona bark contains quinine—a substance still used to treat malaria—but at the time, no one knew why the bark was effective." This information best supports answer choice C.

34. The best answer is J. The passage states, "Modern scientists have been unable to find any evidence to support the theory of molecular memory. In fact, the idea that diluting a substance makes it stronger runs against the principles of chemistry and physics." Based on this information, one can infer that modern doctors are unable to find evidence to support homeopathic treatment because the theory of molecular memory is contrary to the principles of chemistry and physics.

35. The best answer is C. The last paragraph of the passage states, "Today, homeopathy is generally disregarded by many conventional medical practitioners. Homeopathic practitioners are frequently termed quacks by conventional scientists. Nevertheless, homeopathy remains extremely popular both in the United States and abroad. In European countries such as France and England, conventional doctors frequently prescribe homeopathic treatments for common illnesses, such as colds and flu. Pharmacists who are trained to answer questions about the homeopathic treatments' use and desired effects then fill the prescriptions." Based on this information, one can conclude that, although there is a lack of clinical data, homeopathy remains a popular form of treatment, even among doctors and pharmacists.

36. The best answer is H. As is explained by the passage, "When a person became ill, doctors believed it was because one or more of the humors had come out of balance. Some of the best treatments were thought to be bloodletting and purgation—with the assumption being that these treatments would effectively drain off the excess humors." This information best supports answer choice F.

37. The best answer is B. As described in the second paragraph of the passage, "… Sanguine, or passionate, persons were hot and wet."

38. The best answer is J. The fourth paragraph of the passage states that, "Perhaps not surprisingly, Hahnemann's new field of homeopathy (i.e. "similar suffering") was met with considerable resistance from doctors comfortable with their usual practices." As it is used in this sentence, one can infer that *practices* most closely means the various treatments employed by doctors of Hahnemann's era. This information best supports answer choice J.

39. The best answer is B. The fourth paragraph of the passage explains, "Hahnemann further hypothesized that, while undiluted substances would only worsen symptoms in the sick, heavily diluted substances could be effective for a cure." This information best supports answer choice B.

40. The best answer is H. The last paragraph of the passage explains that homeopathy is a controversial form of medical treatment. "Nevertheless, homeopathy remains extremely popular both in the United States and abroad." This information best supports answer choice H.

Science Reasoning Test Explanations

PASSAGE I

1. **The correct answer is A.** In Figure 2, lake clay is indicated by the horizontal stripes. Of the 4 areas listed (Genola, Site 1, Site 2, and Provo) the lake clay deposit is thinnest at Genola.

2. **The correct answer is H.** The passage states that Lake Bonneville was formed "about 32,000 years ago" and that "it existed until about 16,800 years ago." Therefore, you can determine that it existed in its entirety for approximately 32,000 − 16,800 = 15,200 years.

3. **The correct answer is D.** To answer this question, look at the relationship between the area marked by horizontal stripes (lake clay) and the area marked by vertical stripes (bedrock). As the lake clay gets thicker, the bedrock below it gets thinner. Therefore, the thickness of the bedrock decreases.

4. **The correct answer is G.** According to Figure 2, both Site 1 and Site 2 are situated atop the lake clay layer, with Site 2 at a slightly higher elevation than Site 1. The only graph that shows Site 2 at a higher elevation is answer choice G.

5. **The correct answer is A.** The passage states that, "Over time, increasing temperatures in North America caused the lake to begin drying up, leaving Great Salt Lake … the salinity levels are higher than normal." You can deduce from this statement that rising temperatures led to increased water evaporation, which led to increased salinity levels. Therefore, a decrease in temperature and an increase in freshwater rainfall would likely lead to *decreased* salinity levels in Great Salt Lake.

PASSAGE II

6. **The correct answer is H.** You are given that "The order *Lepidoptera* includes butterflies and moths. Table 1 is a key for identifying some Lepidoptera in North America." Because butterflies and moths are not birds, reptiles, or mammals, they must be insects. Even though you might never have seen the word *Lepidoptera* before, you can still correctly answer the question because the term is defined for you.

7. **The correct answer is B.** To answer this question, find *Agraulis vanillae* on Table 1. According to Table 1, the upper side of *Agraulis vanillae* wings is orange with black markings. This species does not have a fuzzy body—Table 1 indicates that you should skip to Step 3 if the insect has a fuzzy body, bypassing *Agraulis vanillae* completely. Likewise, the features mentioned in answer choices C and D are located after *Agraulis vanillae* on the table.

8. **The correct answer is H.** To answer this question, find the listed traits of lepidoptera *Y* in Table 2, and apply them to the steps in Table 1. Table 2 indicates that lepidoptera *Y* has a slim body; therefore, go to Step 2 and eliminate answer choice F. Table 2 indicates that lepidoptera *Y* has yellow wings with markings on the upper side, so go to Step 4 and eliminate answer choice G. Table 2 indicates that lepidoptera *Y* has silver markings, so go to Step 6. Because the silver markings of *Speyeyria coronis* are round and elongated and lepidoptera *Y* has triangular shaped silver markings, you can deduce that lepidoptera *Y* is not a *Speyeyria coronis*.

9. **The correct answer is C.** To answer this question, evaluate each of the answer choices and eliminate those that are incorrect:
   Answer choice A: Step 1—According to Table 2, both *W* and *Z* have a fuzzy body; eliminate answer choice F.
   Answer choice B: Step 3—According to Table 2, both *W* and *Z* have wings with yellow on the upper side; eliminate answer choice G.
   Answer choice C: Step 7—According to Table 2, *W* has prominent brown bands on its underside, and *Z* has no prominent markings on its underside. Therefore, Step 7 is where they first diverge.

10. **The correct answer is F.** To answer this question, first notice that *Automeris io* and *Antheraea polyphemus* are both listed in Step 8. According to Table 1, to get to Step 8 the lepidoptera must have pronounced spots on their hindwings. Note that the lepidoptera in the question have different wingspans, so answer choice H is incorrect. Also, by moving directly to Step 8, Steps 6 and 7 are skipped, so eliminate answer choice G. The only way to get to Step 5 is to have a fuzzy body, so eliminate answer choice J.

PASSAGE III

11. **The correct answer is C.** According to Figure 1, the dotted line represents a water temperature of 40°. Even though 3.0 seconds is not listed on the graph, you can determine that it is halfway between 2.0 and 4.0 seconds; at 3.0 seconds, the dotted line is at approximately 36°.

12. **The correct answer is F.** According to Figure 2, the dotted line represents an air temperature of 30°. During the interval from 0–100 seconds, the dotted

line moves rapidly down and to the right, showing a significant drop in temperature. The dotted line begins to level off shortly after 100 seconds, indicating a stable temperature from about 200 seconds forward.

13. **The correct answer is B.** To answer this question, look at Figure 1. Between 0 and 2 seconds, the thermometer in the 40° water registered a temperature increase from 20° to 30°. Therefore, during the 2-second interval the temperature increased 10°, and the increase in temperature per second was 5°.

14. **The correct answer is J.** You are given that a thermometer at an initial temperature of 60° was placed in air samples at 30°, 40°, and 50°. According to Figure 2, it took approximately 200 seconds for the temperature to stabilize at 30°. You can hypothesize that, if the 60° thermometer was placed in a colder air sample, it would take more than 200 seconds for the temperature to stabilize.

15. **The correct answer is A.** To answer this question, look at Figure 1. The dashed line represents 60° and the solid line represents 50°. The dashed line is always higher than the solid line, so eliminate answer choices C and D. According to the figure, at 4.0 seconds the thermometer in the 60° water registered a temperature of about 48°, which is higher than the temperature recorded after 2.0 seconds (about 38°).

PASSAGE IV

16. **The correct answer is H.** Scientist 1 supports the use of placebos in clinical research. According to Scientist 1, placebos "help address the issue of mind over matter, which is an important issue when working towards treatment for a particular illness." In addition, Scientist 1 states that, "… the possibility of receiving treatment often provides patients with a psychological boost." These statements best support answer choice H.

17. **The correct answer is B.** Scientist 1 states that, "… the possibility of receiving treatment often provides patients with a psychological boost. The use of placebos addresses the question of whether a person's positive attitude may be important in recovery from illness." These statements best support answer choice B.

18. **The correct answer is G.** Both Scientists address the use of placebos in clinical research; Scientist 1 supports the use of placebos while Scientist 2 argues against the use of placebos. Both Scientists discuss the placebo effect as a possible result of using placebos in clinical research. Therefore, the only statement that both Scientists would agree with is answer choice G.

19. **The correct answer is C.** Scientist 2 states that, "Due to its capability to skew research results, the placebo shouldn't be used in clinical research."

20. **The correct answer is J.** Scientist 2 believes that the use of placebos encourages deception in the doctor-patient relationship, which suggests that the use of placebos would lead to less positive communication between the patients and their doctors. Because the finding indicates that the opposite is true—in fact, that the patients were very trusting of their doctors and freely communicated with them—Scientist 2's viewpoint would be weakened.

21. **The correct answer is A.** Scientist 1 states that placebos "establish a control group for the test treatment in question …" If an observation were made suggesting that patients in a control group—the group receiving the placebo—did not recover as quickly as patients who received the medical treatment, Scientist 2's viewpoint against the use of placebos would be weakened. According to Scientist 2, "placebo-related changes could be over-estimated. Different illnesses, by definition, will react differently to the placebo." The observation in answer choice A would support using placebos to help determine the efficacy of a given medical treatment.

22. **The correct answer is J.** Because Scientist 1 believes that using placebos is beneficial in clinical research, Scientist 1 must assume that the use of placebos is safe for most patients. None of the other answer choices are supported by the passage.

PASSAGE V

23. **The correct answer is B.** According to Figure 2, tap water has a viscosity of approximately 1.0, while Motor Oil A has a viscosity of approximately 20.0. Additionally, the results of Experiments 1 and 2 show that the general flow rate of Motor Oil A is lower than that of tap water.

24. **The correct answer is G.** The results of Experiment 1 show that, as the angle of the pipe increases, the velocity of flow increases as well. Therefore, there is a direct relationship.

25. **The correct answer is B.** According to the passage, "Laminar flow is common only in cases in which the flow channel is relatively small, the fluid

is moving slowly, and its *viscosity* (the degree to which a fluid resists flow under an applied force) is relatively high." This information best supports answer choice B, because the pipe diameter is small, the fluid is moving slowly, and Motor Oil B has the highest viscosity of the liquids tested.

26. **The correct answer is C.** According to the passage, Motor Oil B has the highest viscosity, followed by Motor Oil A, followed by tap water. The tables indicate that Motor Oil B has the slowest flow rates, Motor Oil A has slightly faster flow rates, and tap water has the fastest flow rates, overall. This information best supports answer choice C.

27. **The correct answer is A.** According to Table 1, the flow rate of a pipe at a 30° angle with a diameter of 0.25 was 4.0 ft/s, while the flow rate of a pipe at a 30° angle with a diameter of 0.50 was 9.0 ft/s. Therefore, a pipe with a diameter between 0.25 and 0.50 would likely have a flow rate between 4.0 ft/s and 9.0 ft/s. Only answer choice A fits within these parameters.

28. **The correct answer is H.** The three experiments varied only in the type of liquid being used—tap water, Motor Oil A, or Motor Oil B. All of the other factors listed were identical in each experiment.

PASSAGE VI

29. **The correct answer is C.** Both Table 1 and Figure 1 indicate that, as the concentration of sulfite is doubled, the corrected absorbance approximately doubles; 2 is 1 doubled, and 0.302 is approximately 0.153 doubled.

30. **The correct answer is J.** To answer this question, look at Table 2. You are given that dried pineapple measures 0.603, which is higher than prunes; therefore, eliminate answer choices F and H. The order of decreasing concentration is: dried apricots at 0.774; dried pineapple at 0.603; and prunes at 0.562.

31. **The correct answer is B.** According to Table 1, a sulfite concentration of 1.5, which is halfway between 1.0 and 2.0, would have a corrected absorbance about halfway between 0.153 and 0.302.

32. **The correct answer is J.** According to the passage, a colorimeter is a device that measures how much light of a selected wavelength is absorbed. Because different colors have different wavelengths, the colorimeter needs to be adjusted to read at its maximum capacity. The other answer choices are not supported by the passage.

33. **The correct answer is A.** According to Table 1, all of the measured absorbances are higher than the corrected absorbances for the fruits tested. Likewise, the sulfite concentration has a direct relationship with the corrected absorbance in both Table 1 and Table 2. This information best supports answer choice A.

34. **The correct answer is F.** You are given that *additional* contents in the fruits are absorbing light, which means that the apparent absorbance reading will be higher.

PASSAGE VII

35. **The correct answer is A.** The passage states that the molar heat of fusion (ΔH_{fus}) of water is 6.02. This value is less than the molar heat of fusion of calcium.

36. **The correct answer is H.** According to the passage, molar heat of fusion (ΔH_{fus}) is "the amount of heat necessary to melt (or freeze) 1.00 mole of a substance at its melting point at a constant pressure." The table indicates that the molar heat of fusion for iron is 13.8 kJ/mol.

37. **The correct answer is C.** According to the passage, molar heat of fusion (ΔH_{fus}) is "the amount of heat necessary to melt (or freeze) 1.00 mole of a substance at its melting point at a constant pressure." The table shows that molar heat of fusion and melting point both increase for calcium, silver, and iron, but the melting point decreases as the molar heat of fusion continues to increase for nickel. This best supports answer choice C.

38. **The correct answer is F.** The table shows that, in general, higher boiling points result in a higher molar heat of fusion. Therefore, because the boiling point for potassium is lower than the boiling point for calcium, it is likely that the molar heat of fusion for potassium will be lower than the molar heat of fusion for calcium.

39. **The correct answer is C.** You are given that the molar heat of fusion is directly related to the strength of the forces that hold molecules together; therefore, higher molar heats of fusion will indicate stronger bonds between molecules. Iron has a higher molar heat of fusion than calcium, so the

forces holding molecules together will be stronger in iron than in calcium.

40. The correct answer is H. To answer this question, compare the boiling points and molar heats of fusion for each of the pairs of elements:

I. Nickel and iron—the boiling point of iron is higher than the boiling point of nickel, but the molar heat of fusion is lower for iron than it is for nickel. Roman numeral I does *not* support the hypothesis, so eliminate answer choices F and J.

II. Water and calcium—the boiling point of water is lower than the boiling point of calcium, and the molar heat of fusion is lower for water than it is for calcium. Roman numeral II supports the hypothesis, so eliminate answer choice G.

The process of elimination leaves you with answer choice H, but if you evaluate Roman numeral III you will see that it also supports the hypothesis.

Writing Test Explanation

Because grading the essay is subjective, we've chosen not to include any "graded" essays here. Your best bet is to have someone you trust, such as your personal tutor, read your essays and give you an honest critique. If you plan on grading your own essays, review the grading criteria and be as honest as possible regarding the structure, development, organization, technique, and appropriateness of your writing. Focus on your weak areas and continue to practice in order to improve your writing skills.

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